An Explainable AI Framework for Early Academic Project Delay Prediction and Dependency-Aware Task Prioritization

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Abstract - Academic projects frequently experience schedule pressure, uneven workload, residual unresolved work packages and tasks, and milestone slippage; however, conventional project-management tools tend to record project progress and status, rather than estimate future risk. Studies in educational analytics and project-risk management have developed useful predictive techniques to forecast project outcomes in general, but they have not specifically addressed the combination of academic-project-specific early warning, interpretable risk evidence, and task-level intervention capability. We develop an integrated decision-support framework that transforms project activity data into progress, schedule, workload, and dependency, and activity features; uses these features to estimate a High, Medium, or Low milestone-delay risk; provides interpretability by explaining the model output; and then proposes a task ranking framework that identifies numerous tasks that will block future milestones and places them near the top of the ranking if the deadline is near or the downstream impact is large. In this incremental, early examination, we regenerate a version of the proposed student project dataset of 1,200 milestone-level observations that represent 300 simulated project teams; we then provide a brief model comparison that shows that Logistic Regression performs best with an F1-score of 0.721 and a macro ROC-AUC score of 0.879, while the Random Forest model performs well with an F1-score of 0.687; we then conduct simulations of risk detection, and we demonstrate that our simple deadline-based baseline yields a macro F1-score of 0.610 and that using the full feature set resulted in an F1-score of 0.687; we also show that the Random Forest model finds 63.0% of the simulated High-risk milestones after observing only 30% of the simulated project, and we simulate task prioritization for a more than doubled dataset and show that our method places 92.2% of the blocking tasks in the top 20% of the priority ranking compared to 17.6% by the baseline method that considers only the deadline. These findings provide an initial validation of the framework: the labels and task relationships are simulated, generalization to a real academic project remains untested, and this framework is intended to support faculty and student decisions rather than to autonomously evaluate students.

Index Terms - academic project management, early warning, project risk prediction, explainable artificial intelligence, task prioritization, dependency-aware scheduling, educational analytics.

I. INTRODUCTION

Academic projects diverge from typical task-management contexts in that project progress is constrained by semester schedules, faculty reviews, shared resources, and milestone-based evaluation. A typical project undergoes topic selection, literature review, design, implementation, testing, documentation, and presentation. A delay in one stage can therefore propagate into later activities, particularly when tasks are connected by dependencies. In practice, teams often maintain this information through meetings, spreadsheets, messaging platforms, or project boards. Such tools can record what has already happened, but provide limited support for estimating what is likely to happen.

The distinction matters because project risk is not determined by the number of unfinished tasks alone. A team may have several pending tasks while remaining healthy if those tasks are low-impact and have sufficient schedule slack. Conversely, a single incomplete integration task can place a project under substantial pressure when testing and documentation depend on it. Traditional deadline ordering does not capture this difference. A more useful early-warning mechanism should combine the current state of the project with the relationships among its activities.

Predictive analytics offers one approach to this problem. Educational data-mining research has demonstrated that machine-learning models can identify patterns associated with student performance and can support early intervention [2]–[4]. Project-risk research has also examined predictive models for identifying schedule and performance risks, increasingly paired with interpretable explanations [5]–[7]. These two research streams, however, target different objects: educational analytics generally predict learner outcomes, while project-risk studies typically target industrial projects. Academic project teams occupy an intermediate space in which project activity, student workload, milestone progress, and faculty supervision intertwine.

Prediction alone is not enough for intervention. A warning such as "High risk" does not tell a student or faculty guide what evidence produced the warning or which task should be addressed first. Explainability methods can expose the features associated with a model output [8]–[10], while dependency-aware ranking can translate project state into a concrete order of work. The intended chain is therefore not simply activity → prediction, but activity → prediction → explanation → prioritization → intervention support.

This paper examines that integrated chain in the academic-project setting. The contribution is deliberately narrower than a claim of novelty for machine-learning project-risk prediction. The focus is an academic-project-oriented framework that connects milestone-delay risk estimation with interpretable evidence and dependency-aware task prioritization. Because institutional project records were unavailable for this study, the experimental evidence is drawn upon a newly regenerated synthetic dataset and a separate synthetic task graph. The experiments are therefore treated as controlled validation of the method rather than as evidence of real-world deployment performance.

II. RESEARCH QUESTIONS AND CONTRIBUTIONS

A. Research Questions

- RQ1: Can progress, deadline, workload, dependency, and activity features enhance simulated academic project milestone-delay risk prediction beyond the baseline of a deadline-based threshold?
- RQ2: At what stage in the project timeline can the framework detect simulated High-risk milestones, assuming only information available through the prediction point is leveraged?
- RQ3: Which project features dominate the model predictions, and is such contribution amenable to an interpretable presentation?
- RQ4: Can dependency-aware task prioritization outperform deadline-only ranking in identifying schedule-critical blocking tasks?

B. Research Contributions

- An academic-project oriented feature-engineering formulation, combining progress, schedule, workload, dependency, and activity signals into a unified milestone-delay risk estimation.
- An early-warning classification workflow comparing across multiple supervised learning approaches on a project-grouped experimental split.
- A dependency-aware task prioritization approach, incorporating downstream impact weights in addition to the next-deadline time constraint into the task ranking.
- An SHAP-based explainability layer for the selected tree-based model, as well as the methodological differentiation of model contribution and causal influence.
- An integrated decision-support architecture, linking risk estimation, explanation, task ranking, and traceable actionable recommendations.

III. RELATED WORK

A. Academic Project Management

Academic project management has focused on the structured documentation, scheduling, roles, and supervision. Styk, Liszcz, and Drobek demonstrate the student-project workflow that was based on simple project documents, emphasizing the value of systematic project records in an academic setting [1]. Such systems do target organization and traceability, but their main purpose is management, not predictive early warning. The current work retains these management functions as an information source for the predictive layer, not as the research contribution.

B. Educational Predictive Analytics

Educational data mining has extensively studied student performance prediction and early intervention. Alalawi et al. reviewed 162 studies and found classification to be the dominant approach, using methods such as decision trees, random forests, and neural networks [2]. Cano and Leonard developed an interpretable multiview early-warning system, demonstrating the value of combining prediction with interpretability [3]. Çırak, Akıllı, and Ekinci evaluated an early-warning system using 11,698 first-year students and multiple classifiers [4]. However, these studies focus on student-level outcomes rather than project milestones and task dependencies.

C. Project Risk Prediction and Explainability

Machine learning has also been applied to project-risk assessment, particularly in the construction and other industrial environments. Badhon et al. explored explainable project-risk assessment and later proposed multi-module frameworks that explicitly addressed risk-factor relationships and interpretability [5], [6]. Cho et al. combined delay prediction, risk prioritization, and SHAP-based interpretation in highway earthwork projects [7]. These works are closely related to the predictive-to-decision-support direction of the current work, but the underlying context, data structure, and intervention unit differ. The proposed framework addresses the academic project team and its milestones as the decision context.

D. Task Prioritization and Dependencies

Task prioritization is often driven by urgency, importance, and schedule constraints, while dependence-aware scheduling recognizes that the impact of an activity partially depends on the following work. In the current framework, dependence is explicitly represented because a task blocking several downstream activities can be more consequential than another task with an earlier nominal deadline. This design is also consistent with recent project-risk research that has transitioned from prediction to risk prioritization and action support [7]. The difference lies in the use of task-level downstream impact within an academic-project workflow.

E. Explainable Artificial Intelligence

Explainable artificial intelligence provides methods to examine how model inputs affect predictions. Adadi and Berrada surveyed XAI methods and emphasized the importance of transparency when complex models are used in decision-support settings. SHAP formulates feature attribution as additive Shapley-based explanations [9], while LIME provides local surrogate explanations around individual predictions [10]. The current work uses SHAP for the Random Forest model because the objective is to provide both global feature importance and instance-level evidence, not to replace the underlying predictive model.

F. Research Gap

The literature therefore covers three relevant areas: academic analytics and early warning, project-risk prediction, and explainable decision support. What remains comparatively unexplored is their combination around an academic project as the unit of analysis, particularly when the prediction is tied to dependence-aware task prioritization. The gap addressed here is therefore not the absence of machine learning in project management, but the lack of an integrated academic-project workflow in which structured project activity is transformed into an early risk signal, explained, and translated into task-level intervention support.

TABLE I — COMPARISON OF EXISTING APPROACHES

Study	Academic context	Risk prediction	Early warning	Explain-ability	Task priority	Dependency awareness
Styk et al. [1]	Yes	No	No	No	No	No
Alalawi et al. [2]	Yes	Yes*	Yes	Limited	No	No
Cano & Leonard [3]	Yes	Yes*	Yes	Yes	No	No
Çırak et al. [4]	Yes	Yes*	Yes	Not central	No	No
Badhon et al. [5], [6]	No†	Yes	Proactive	Yes	Risk-level	Risk-level
Cho et al. [7]	No†	Yes	Yes	Yes	Yes	Not explicit
Proposed framework	Yes	Yes	Yes	Yes	Yes	Yes

** Educational outcome prediction rather than project-milestone delay prediction. † Industrial project context.*

IV. PROPOSED FRAMEWORK

A. System Overview

The framework is organized as a decision-support pipeline rather than as a conventional project-management application. The management platform supplies structured observations; the predictive layer transforms those observations into risk estimates; the explanation layer identifies influential features; and the prioritization layer ranks tasks according to schedule urgency and downstream impact. The final output is a traceable recommendation that preserves the connection between evidence and action.

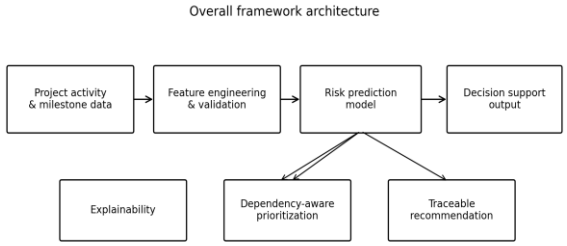


Fig. 1. Overall architecture of the academic-project decision-support framework.

B. Data Flow

At a prediction point, the system first validates the current project snapshot and prunes out information that would only be available after the prediction point. The resulting feature vector is then passed to the trained classifier. The model will return class probabilities and a risk category. SHAP values are then computed for the selected Random Forest model. Meanwhile, the task-prioritization module will normalize task-level features and produce a priority score. Finally, the recommendation engine will combine the predicted risk, evidence in support of the prediction, and highest priority tasks and create a succinct intervention statement.

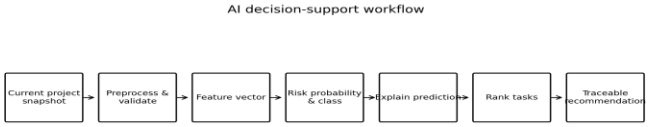


Fig. 2. End-to-end workflow from current project activity to decision-support output.

C. Risk Prediction Formulation

For each project-milestone observation i, the observable input vector is defined as:

$$X_i = [P_i, D_i, W_i, Dep_i, A_i]$$

where P denotes progress, D denotes schedule and deadline indicators, W denotes workload, Dep denotes dependency risk, and A denotes recent activity. The model estimates a probability distribution over three simulated risk classes:

$$\hat{Y}_i = f(X_i), \quad P(Y_i \mid X_i), \quad Y_i \in \{Low, Medium, High\}.$$

The synthetic target is a surrogate for future delay propensity rather than an observed institutional outcome. It is generated from a documented latent risk score in Section V. This distinction is important: the experiment evaluates whether the feature representation and prediction pipeline can recover simulated risk structure; it does not establish that the model predicts real academic project delays.

D. Dependency-Aware Task Prioritization

For a task T_i, the method normalizes six task characteristics: deadline urgency D_i, dependency impact Dep_i, project-risk contribution R_i, task importance I_i, delay severity L_i, and effort priority E_i. The priority score is:

$$Priority(T_i) = 0.25D_i + 0.45Dep_i + 0.10R_i + 0.05I_i + 0.10L_i + 0.05E_i.$$

The weights sum to one and were fixed before the task-level evaluation. Dependency impact receives the largest weight because the principal purpose of the ranking is to surface tasks whose completion controls downstream work. Deadline urgency is the next-largest component because schedule pressure remains important. The remaining terms provide secondary context rather than replacing the dependency signal. A blocking task is defined as a task with at least one downstream task that depends on its completion.

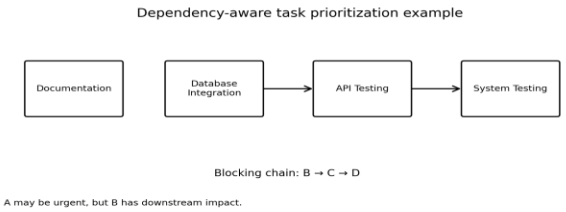


Fig. 3. Example dependency chain illustrating why downstream impact can change task priority.

E. Explainability

SHAP is used to quantify how individual features contribute to the selected Random Forest prediction. Global importance is obtained from the mean absolute SHAP value across the test observations and classes. For an individual observation, positive and negative SHAP contributions indicate movement toward or away from the predicted class. These values describe model behavior, not causality. A feature with a large contribution should therefore be interpreted as evidence used by the model rather than proof that the feature caused the project condition.

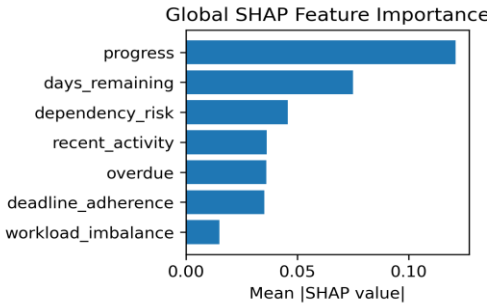


Fig. 4. Global SHAP feature importance for the Random Forest model under the regenerated synthetic evaluation.

F. Recommendation Generation

Recommendations are produced from structured evidence, not directly from unconstrained language- model generation. The rationale follows a logical chain of evidence from four steps: prediction → evidence → priority → recommendation. For instance, a predicted High-risk status combined with low progress, high predicted risk of dependencies, and a blocking integration task leads to a recommendation to address the integration task before lower-risk work. A language model could later summarize the evidence, but such a model would not be part of the algorithm itself and is not required for reproducing the prediction.

G. End-to-End Algorithm

Algorithm 1: Academic Project Risk and Task Prioritization

Input: Current project activity, task graph, trained classifier

1) Validate the current snapshot and remove post-prediction information
2) Calculate progress, schedule, workload, dependency, and activity features
3) Apply trained classifier and calculate probabilities for each class
4) Predict Low, Medium, or High risk based on predicted class
5) Calculate SHAP values for the selected model
6) Normalize urgency, dependency, risk, importance, delay, and effort
8) Find highest-priority blocking or schedule-critical tasks
9) Convert evidence and ranking into traceable recommendation
10) Output risk, evidence, sorted list, and recommendation to the user

V. DATASET AND FEATURE ENGINEERING

A. Synthetic Dataset Construction

No institutional project records were available for this study. A new controlled synthetic dataset was therefore generated to permit reproducible testing of the proposed pipeline. This dataset contains 300 simulated project teams, each of which contributes 4 milestone-level observations, for 1200 observations in total. The random seed is 20260831. Each observation corresponds to the state of one of milestone points in the project lifecycle.

The data generator first assigns project-level tendencies towards workload and dependencies, and then samples milestone-level difficulty, progress, workload, dependency risk, deadline adherence, recent activity level, and overdue-task count, based on these tendencies. The synthetic generator is deliberately designed such that a group of signals jointly contribute to a latent delay-propensity score. The labels are then balanced using the lower and upper tertiles of this score to make the evaluation suitable for testing the pipeline, while avoiding the appearance that the data were collected from real students or an institution.

TABLE II — SYNTHETIC DATASET CHARACTERISTICS

Property	Value
Simulated project teams	300
Milestones per team	4
Total observations	1,200
Random seed	20260831
Training / test split	900 / 300 observations
Grouping rule	No project appears in both sets
Risk classes	Low: 400; Medium: 400; High: 400
Label basis	Tertiles of synthetic latent risk score
Institutional records	Not used

B. Feature Groups

The model uses seven observable variables. Progress is expressed as the percentage of milestone work completed. Overdue is the number of overdue tasks associated with the milestone. Deadline adherence has a value between 0 and 1 that represents recent compliance with planned dates. Workload imbalance is normalized between 0 and 1. Dependency risk is a normalized estimate of unresolved downstream dependency pressure. Recent activity is a normalized activity measure. Days remaining is the estimated schedule distance to the milestone deadline and is used in concert with overdue count in the deadline-based baseline.

Across the complete, synthetic dataset, progress varies between 5 – 100%, the overdue-task count between 0 – 10, deadline adherence between 0 – 1, workload imbalance and dependency risk between 0 – 1, recent activity between 0.06 – 1.00, and days remaining between 1 – 40 days. These ranges describe the generated data rather than observe institutional behavior.

C. Target Generation

A latent risk score S_i is generated for each observation. Let P_i be normalized progress, A_i recent activity, D_i deadline adherence, W_i workload imbalance, Dep_i dependency risk, O_i normalized overdue count, and Q_i the latent milestone difficulty used only by the simulator. The score is:

$$S_i = 0.28(1-P_i) + 0.18(1-D_i) + 0.16W_i + 0.20Dep_i + 0.10(1-A_i) + 0.08O_i + 0.12Q_i + \varepsilon_i.$$

The observed labels are assigned using the empirical tertiles of S. For the regenerated dataset, the lower and upper thresholds are 0.3591 and 0.4735. Scores at or below 0.3591 are Low, scores between 0.3591 and 0.4735 are Medium, and scores above 0.4735 are High. Because Q_i and ε_i are simulator variables, the target represents simulated future delay propensity rather than an observed completion-date outcome. This limitation is carried through the results and discussion.

D. Data Leakage Prevention

Only information that was intended to be available at the time of prediction is included as model features. Information about future completion, future review scores, and the target-generating score itself are excluded from model input. The project identifier is used only for grouping purposes, and never as a predictive feature. Preprocessing and standardization for Logistic Regression is fit on the training portion. The 75/25 split is made at the project level so that a milestone from a single simulated project would not appear in both training and testing. This mitigates a model from learning project-specific patterns from one milestone and subsequently exploiting this information on another milestone from the same project.

VI. EXPERIMENTAL METHODOLOGY

A. Models

Four supervised classifiers were evaluated. Logistic Regression provides a linear and relatively interpretable baseline. Decision Tree captures nonlinear, threshold relationships. Random Forest consists of an

ensemble of randomized decision trees, and is well suited to structured tabular data [11]. XGBoost represents gradient-boosted trees, and provides a strong nonlinear comparator [12], [13]. The experiments are carried out using scikit-learn compatible implementations.

TABLE III — MODEL CONFIGURATION

Model	Important parameters	Training / validation
Logistic Regression	StandardScaler; max_iter=2000	Project-grouped 75/25 holdout
Decision Tree	max_depth=5; min_samples_leaf=8	Project-grouped 75/25 holdout
Random Forest	300 trees; max_depth=8; min_samples_leaf=3	Project-grouped 75/25 holdout
WWAZZXGBoost	``111111	Project-grouped 75/25 holdout

The settings were frozen before the final test evaluation and were not tuned against the test set. The goal of this controlled study was to reproduce the regeneration experiment, not to exhaustively search for hyperparameters.

B. Deadline-Based Baseline

The baseline reflects a standard approach to schedule monitoring. A scalar baseline score is formed by normalized proximity to deadlines and count of overdue tasks with respective weights of the two terms of 0.65 and 0.35 and thresholds derived only from the training portion of the data. This baseline asks whether the extra information about progress, workload, dependencies, and activity provides a valuable signal beyond schedule pressure.

C. Evaluation Metrics

Accuracy, macro-averaged precision, macro-averaged recall, and macro F1-score are reported. Macro-averaging assigns equal weight to each risk class. ROC-AUC is calculated using the one-vs-rest formulation also macro-averaged across the three classes. For the early-warning experiment, High-risk recall is reported separately because the failure to recognize a truly high-risk observation would be more costly to an early-warning system than an isolated error in one of the other classes.

D. Early-Risk Evaluation

The early-risk performance is evaluated at 30%, 50%, 70%, and 90% project completion. At each snapshot, only the information that would have been observable up to that stage was used. Earlier snapshots have noisier estimates of activity and dependency state, whereas later ones contain more information. The target for prediction was the eventual synthetic risk class assigned to the milestone. No target-generating variables were passed to the classifier.v



Fig. 5. Temporal evaluation procedure. Each snapshot uses only information available up to that stage before predicting the eventual simulated risk class.

E. Feature Ablation

A cumulative ablation study evaluates the Random Forest using the deadline signal only, then with progress, workload, dependency, and recent activity features added in succession. We are not seeking an optimal feature ordering (such an ordering likely depends on the particular holdout split), but rather whether the combined representation has value beyond its components, on the same holdout split.

F. Task-Prioritization Evaluation

On the separate synthetic task graph, with 600 total tasks and 102 blocking tasks, the top 20% corresponds to the 120 highest-ranked tasks. The blocking-task recall@20% thus defined is the percentage of the 102 blocking tasks that appear in the top 120 according to each ranking. Four variants of the proposed approach are compared, including deadline only, deadline+dependency, deadline+dependency+risk, and the full score.

VII. RESULTS AND ANALYSIS

A. Model Comparison

TABLE IV — MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.727	0.720	0.723	0.721	0.879
Decision Tree	0.597	0.611	0.596	0.600	0.800
Random Forest	0.693	0.686	0.688	0.687	0.853
XGBoost	0.683	0.674	0.678	0.676	0.856
Deadline-only baseline	0.207	0.211	0.209	0.210	—

The Logistic Regression classifier achieved the highest macro F1-score (0.721) and the highest macro ROC-AUC (0.879) among the tested classifiers. The Random Forest classifier attained 0.687 macro F1 and 0.853 macro ROC-AUC, while the XGBoost and Decision Tree classifiers achieved 0.676 and 0.600 macro F1, respectively. The deadline-only baseline hit 0.210 macro F1. The result provides grounds for asserting that the information encoded in the multi-feature representation is partly predictive of the target beyond the deadline proximity and overdue count in the simulated data-generating process.

The Random Forest classifier was selected for the remaining interpretability experiments despite Logistic Regression's aggregate superiority. The choice was motivated by methodological considerations: the Random Forest is capable of capturing nonlinear dependencies while providing both tree-based feature importance and SHAP explanations. The small performance gap between it and the XGBoost classifier also suggests that the evaluation does not provide sufficient grounds for asserting the existence of a uniquely best model.

B. Early-Risk Detection

TABLE V — EARLY PREDICTION PERFORMANCE

Project stage	Precision	Macro recall	Macro F1	High-risk recall
30%	0.576	0.579	0.574	0.630
50%	0.547	0.557	0.546	0.593
70%	0.599	0.611	0.602	0.694
90%	0.581	0.585	0.581	0.565

At the 30% snapshot, High-risk recall was 0.630. It increased to 0.593 at 50%, 0.694 at 70%, and 0.565 at 90% in the regenerated simulation. The variation shows that early warning quality is not necessarily monotonic with project completion. More information is available later, but the synthetic feature distributions and noise also change across snapshots. The key result for RQ2 is that the model identifies a substantial portion of simulated High-risk observations at the earliest tested stage, while the exact recall should not be interpreted as a real-world early-warning rate.

C. Feature Ablation

TABLE VI — FEATURE ABLATION

Feature configuration	Accuracy	Precision	Recall	F1
Deadline signals	0.620	0.609	0.615	0.610
+ Progress	0.583	0.568	0.577	0.572
+ Workload	0.600	0.582	0.593	0.586
+ Dependency	0.650	0.639	0.645	0.641
+ Activity	0.687	0.677	0.681	0.678
Full feature set	0.693	0.686	0.688	0.687

The complete feature configuration obtained a macro F1 of 0.687 against 0.610 for the deadline-signal baseline. The addition of progress alone decreased the score in this particular simulation while dependency and recent activity led to greater gains. This is a valuable insight because it demonstrates that not all features are individually beneficial, and that the predictiveness comes from their joint representation. The ablation thus yields stronger support to the integrated feature design than a simple comparison of the full versus baseline configurations.

D. Explainability Analysis

The SHAP analysis suggests that progress was the leading contributor to the Random Forest predictions from these features, followed by days remaining and dependency risk. Relative mean absolute SHAP importance was approximately 33.2% for progress, 20.6% for days remaining, 12.6% for dependency risk, 9.9% for recent activity, 9.9% for overdue count, 9.6% for deadline adherence, and 4.1% for workload imbalance. These are descriptive values for the model's allocation of predictive contribution in the simulated data and should not be interpreted as casual effects.

For an individual correctly classified High-risk observation, the strongest positive SHAP contributions came from low progress, schedule pressure represented by the remaining schedule, and lower deadline adherence. Other features favored the opposite direction of prediction. This illustrates the intended value of explainability: a user can review the evidence toward a warning rather than receiving only the class label. The explanation is representative of model behavior, not an automated judgment of the student or team

E. Task Prioritization

TABLE VII — TASK PRIORITIZATION EVALUATION

Ranking method	Blocking-task recall@20%
Deadline only	17.6%
Deadline + dependency	64.7%
Deadline + dependency + risk	91.2%
Full proposed ranking	92.2%

The deadline-only ranking captured 17.6% of the blocking tasks in its top 20%. Adding dependency impact raised the value to 64.7%, and adding project risk increased it to 91.2%. The full ranking reached 92.2%. The result highlights the specific role of dependency information: a task's downstream effect can be more informative for intervention than its own due date. The additional terms provide a small further improvement in this simulation.

F. Error Analysis

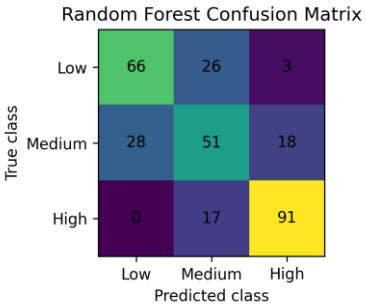


Fig. 6. Random Forest confusion matrix on the project-grouped test set.

The confusion matrix shows that errors remain across adjacent risk classes, which is expected when the three simulated classes are produced from a continuous latent score. The model is more likely to confuse neighboring Low and Medium or Medium and High observations than to make an extreme Low-to-High or High-to-Low transition. This matters for deployment because a false negative on a High-risk project could delay intervention, while a false positive could increase unnecessary supervision. The present simulation does not quantify the operational cost of either error type; that should be studied with institutional stakeholders.

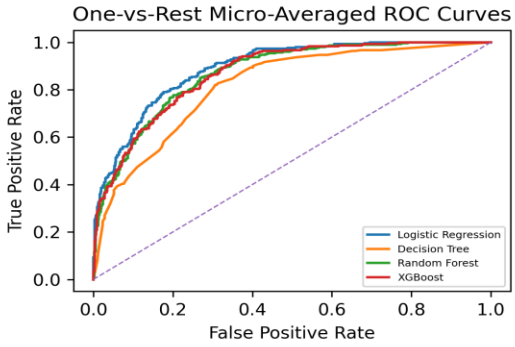


Fig. 7. One-vs-rest micro-averaged ROC curves for the evaluated classifiers.

The ROC analysis is reported as a complementary view of class separation. Logistic Regression produced the highest macro one-vs-rest ROC-AUC in the tabulated evaluation. The ROC result should not be considered sufficient evidence of deployment quality because the test data are synthetic and generated from the same controlled process as the training data.

VIII. PROTOTYPE IMPLEMENTATION

The software prototype is intended to house the research workflow rather than define the research contribution. A Flutter or web client could provide student and faculty views, a Python REST service could manage project data and invoke the prediction component, and a relational database could store projects, tasks, milestones, and reviews. The machine-learning stack employs Python and Pandas, NumPy, scikit-learn, XGBoost, and SHAP. Role-based authentication separates student and faculty access. The implementation can be deployed as a management service and a prediction service to avoid redesigning the application layer when a model is updated.

The application may also expose a project-health indicator in the guise of a compact dashboard summary. This indicator is presented as a presentation feature, not a research metric, as no independently validated health-score equation has been established in this study.

IX. ETHICAL, PRIVACY, AND SECURITY CONSIDERATIONS

Academic risk prediction concerns students and requires a clear separation between project monitoring and student evaluation. The risk output should describe project conditions rather than label an individual as capable or incapable. Faculty members should retain responsibility for intervention decisions, and students should be able to inspect the principal evidence behind a warning.

A real deployment should minimize collected data, remove personally identifiable information from research datasets, restrict access to faculty evaluations, use secure authentication and encrypted communication, and avoid sensitive personal attributes unless their use is justified by a documented institutional purpose. The predictive model should not be used for punitive decisions such as grading, ranking students, or denying academic opportunities. The current synthetic evaluation is not a human-subject study and should not be presented as such.

X. LIMITATIONS

The most important limitation is the synthetic nature of the evaluation. Although the regenerated dataset is reproducible and the experiments are genuinely rerun, the generator cannot reproduce the full variability of real student teams, project domains, faculty practices, or institutional schedules. The reported performance therefore demonstrates feasibility under a controlled simulation rather than generalization to academic institutions.

The effective number of independent projects is 300 even though the dataset contains 1,200 milestone observations. The observations are therefore not independent in the same way as 1,200 unrelated projects. The model also uses correlated schedule features such as progress and days remaining, which may affect feature attribution. Concept drift across semesters, changes in project types, and differences between software, hardware, research, and interdisciplinary projects may further reduce transferability.

Finally, model explanations can be misunderstood as causal statements, and users may over-rely on a risk score. The framework is consequently intended as decision support. Real institutional validation, calibration analysis, and a study of intervention outcomes are required before any operational use.

XI. FUTURE WORK

The next stage is to collect anonymized, institutionally approved project histories covering multiple semesters and project domains. Real milestone completion dates should replace synthetic labels, and the label definition should be based on a pre-registered delay criterion. Cross-semester and cross-department evaluation can then test generalization. Temporal models can be investigated to capture changes in project behavior, while graph-based models can represent task dependencies more explicitly.

Future evaluation should also examine calibration, cost-sensitive errors, and intervention effectiveness. An ablation study on real data can determine whether dependency information consistently adds value. Faculty and student feedback can be used to evaluate whether explanations are understandable and whether recommendations lead to measurable changes in project progress. If a language model is later introduced, it should remain downstream of the deterministic prediction and prioritization components.

XII. CONCLUSION

This paper presented an explainable decision-support framework for early academic project delay-risk analysis and dependency-aware task prioritization. The framework treats project activity as structured evidence rather than as a collection of isolated task records. Progress, schedule, workload, dependency, and activity features are transformed into a risk estimate; SHAP provides an interpretable view of the model output; and a dependency-aware ranking translates the prediction into an actionable task order.

Under the regenerated synthetic evaluation, Logistic Regression achieved the best aggregate classification performance, while Random Forest provided a useful nonlinear model for explanation. The full feature

configuration improved macro F1 over the deadline-signal baseline, and the early-risk experiment indicated that meaningful High-risk recall was possible at the 30% project stage. The task experiment further showed that dependency information substantially improved the retrieval of blocking tasks compared with deadline-only ranking.

These findings answer the research questions only within the controlled simulation. They provide evidence that the integrated framework is technically feasible and that the feature and prioritization design can add information beyond simple deadline monitoring. They do not establish performance on real academic projects, nor do they justify automated evaluation of students. The principal next step is therefore real institutional validation using anonymized project histories, followed by calibration, intervention, and usability studies.

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